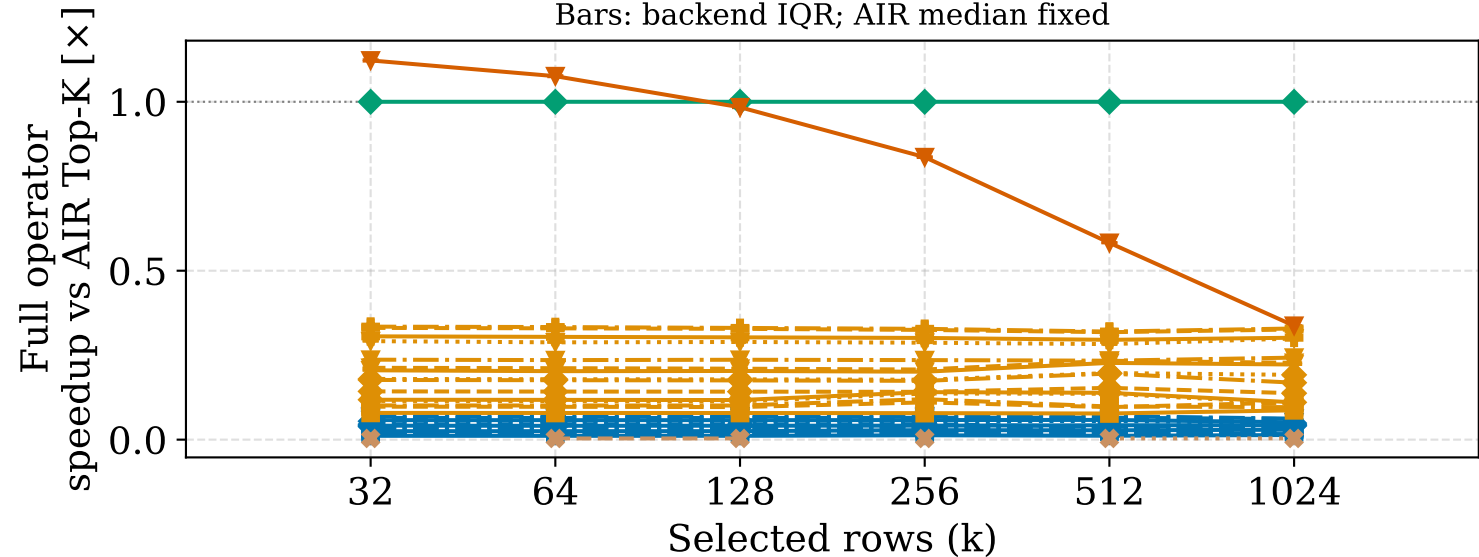


59,986,052 rows; input 2749d31664; split degree=8

AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

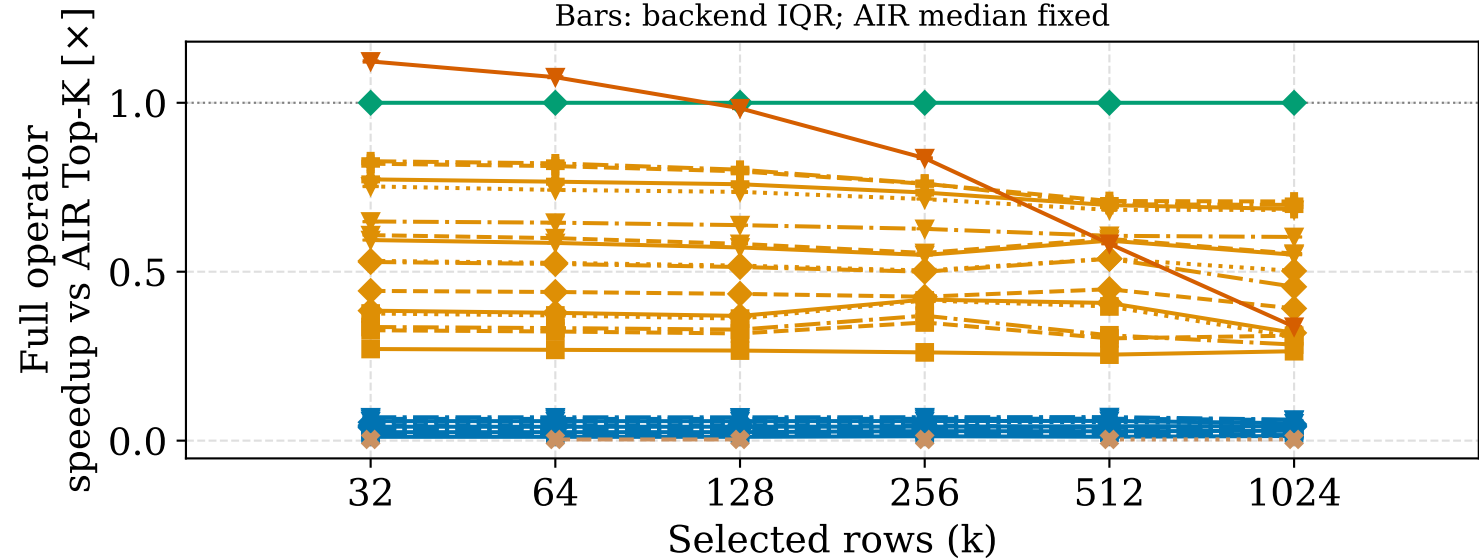


- |                                             |                                              |
|---------------------------------------------|----------------------------------------------|
| bits (degree=1, block=128, items=4)         | bits (split) (degree=8, block=128, items=13) |
| bits (degree=1, block=128, items=7)         | bits (split) (degree=8, block=128, items=16) |
| bits (degree=1, block=128, items=8)         | bits (split) (degree=8, block=256, items=4)  |
| bits (degree=1, block=128, items=13)        | bits (split) (degree=8, block=256, items=7)  |
| bits (degree=1, block=128, items=16)        | bits (split) (degree=8, block=256, items=8)  |
| bits (degree=1, block=256, items=4)         | bits (split) (degree=8, block=256, items=13) |
| bits (degree=1, block=256, items=7)         | bits (split) (degree=8, block=256, items=16) |
| bits (degree=1, block=256, items=8)         | bits (split) (degree=8, block=512, items=4)  |
| bits (degree=1, block=256, items=13)        | bits (split) (degree=8, block=512, items=7)  |
| bits (degree=1, block=256, items=16)        | bits (split) (degree=8, block=512, items=8)  |
| bits (degree=1, block=512, items=4)         | bits (split) (degree=8, block=512, items=13) |
| bits (degree=1, block=512, items=7)         | bits (split) (degree=8, block=512, items=16) |
| bits (degree=1, block=512, items=8)         | AIR Top-K (degree=1, block=512, items=0)     |
| bits (degree=1, block=512, items=13)        | GridSelect (degree=1, block=0, items=0)      |
| bits (degree=1, block=512, items=16)        | BlockSelect (degree=1, block=128, items=2)   |
| bits (split) (degree=8, block=128, items=4) | BlockSelect (degree=1, block=128, items=3)   |
| bits (split) (degree=8, block=128, items=7) | BlockSelect (degree=1, block=128, items=4)   |
| bits (split) (degree=8, block=128, items=8) | BlockSelect (degree=1, block=128, items=8)   |

59,986,052 rows; input 2749d31664; split degree=32

AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

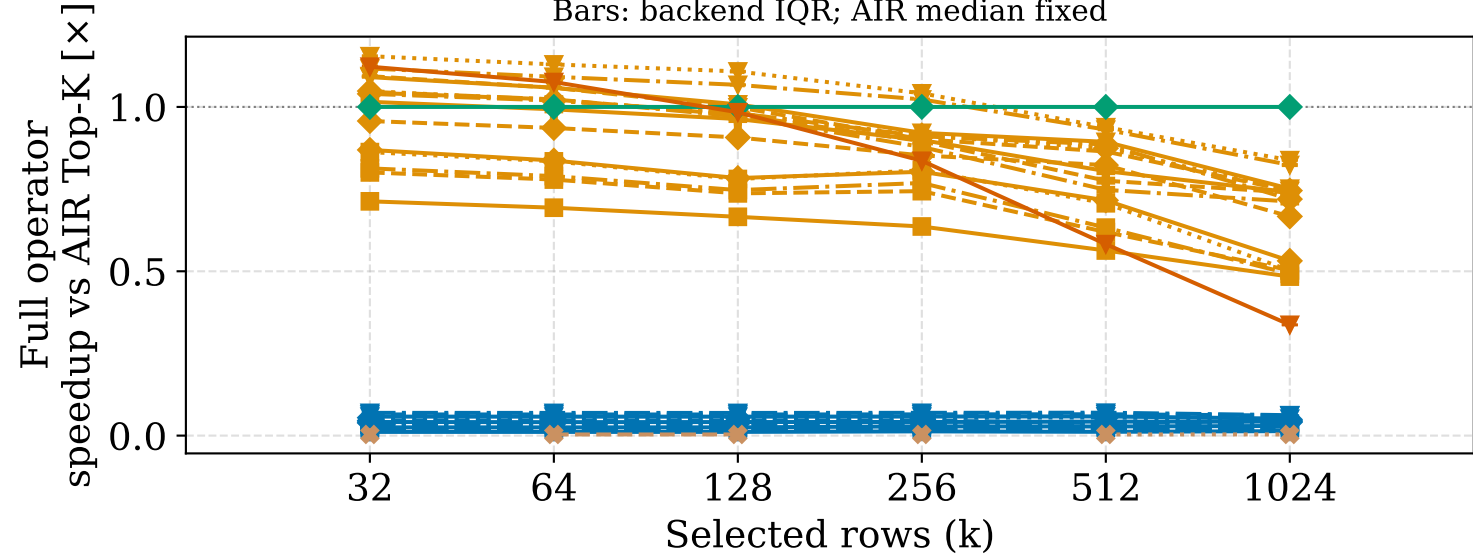


- |                                              |                                               |
|----------------------------------------------|-----------------------------------------------|
| bits (degree=1, block=128, items=4)          | bits (split) (degree=32, block=128, items=13) |
| bits (degree=1, block=128, items=7)          | bits (split) (degree=32, block=128, items=16) |
| bits (degree=1, block=128, items=8)          | bits (split) (degree=32, block=256, items=4)  |
| bits (degree=1, block=128, items=13)         | bits (split) (degree=32, block=256, items=7)  |
| bits (degree=1, block=128, items=16)         | bits (split) (degree=32, block=256, items=8)  |
| bits (degree=1, block=256, items=4)          | bits (split) (degree=32, block=256, items=13) |
| bits (degree=1, block=256, items=7)          | bits (split) (degree=32, block=256, items=16) |
| bits (degree=1, block=256, items=8)          | bits (split) (degree=32, block=512, items=4)  |
| bits (degree=1, block=256, items=13)         | bits (split) (degree=32, block=512, items=7)  |
| bits (degree=1, block=256, items=16)         | bits (split) (degree=32, block=512, items=8)  |
| bits (degree=1, block=512, items=4)          | bits (split) (degree=32, block=512, items=13) |
| bits (degree=1, block=512, items=7)          | bits (split) (degree=32, block=512, items=16) |
| bits (degree=1, block=512, items=8)          | AIR Top-K (degree=1, block=512, items=0)      |
| bits (degree=1, block=512, items=13)         | GridSelect (degree=1, block=0, items=0)       |
| bits (degree=1, block=512, items=16)         | BlockSelect (degree=1, block=128, items=2)    |
| bits (split) (degree=32, block=128, items=4) | BlockSelect (degree=1, block=128, items=3)    |
| bits (split) (degree=32, block=128, items=7) | BlockSelect (degree=1, block=128, items=4)    |
| bits (split) (degree=32, block=128, items=8) | BlockSelect (degree=1, block=128, items=8)    |

59,986,052 rows; input 2749d31664; split degree=128

AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

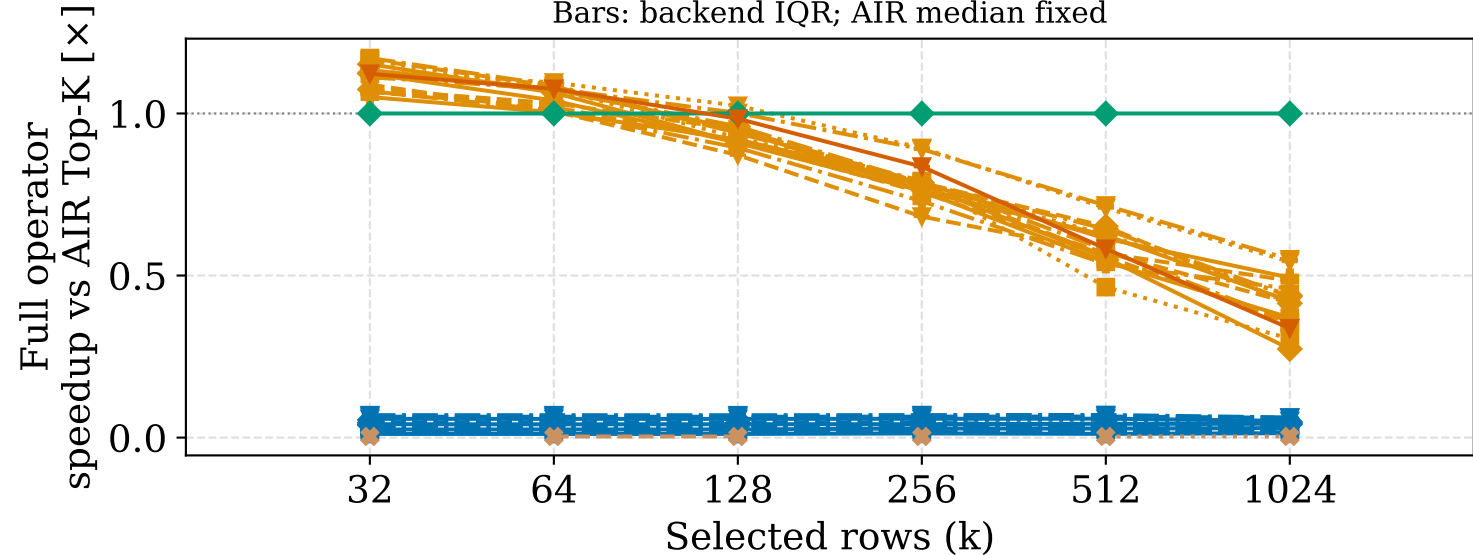


- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=128, block=128, items=4)
- bits (split) (degree=128, block=128, items=7)
- bits (split) (degree=128, block=128, items=8)
- bits (split) (degree=128, block=256, items=4)
- bits (split) (degree=128, block=256, items=7)
- bits (split) (degree=128, block=256, items=8)
- bits (split) (degree=128, block=512, items=4)
- bits (split) (degree=128, block=512, items=7)
- bits (split) (degree=128, block=512, items=8)
- bits (split) (degree=128, block=512, items=13)
- bits (split) (degree=128, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)

59,986,052 rows; input 2749d31664; split degree=512

AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed



- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=512, block=128, items=4)
- bits (split) (degree=512, block=128, items=7)
- bits (split) (degree=512, block=128, items=8)
- bits (split) (degree=512, block=256, items=4)
- bits (split) (degree=512, block=256, items=7)
- bits (split) (degree=512, block=256, items=8)
- bits (split) (degree=512, block=256, items=13)
- bits (split) (degree=512, block=256, items=16)
- bits (split) (degree=512, block=512, items=4)
- bits (split) (degree=512, block=512, items=7)
- bits (split) (degree=512, block=512, items=8)
- bits (split) (degree=512, block=512, items=13)
- bits (split) (degree=512, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)